HYPERC RESEARCH | WORKING PAPER

Computable Markets

Business Menus, Sales Event Tapes, and Cross-Market Profit-Directed Learning

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An operator-relative theory and data contract for learning executable economic decisions from business telemetry

Abstract

This paper introduces and formalizes the **computable market** as an operator-relative category of economic environment: a market in which feasible actions can be represented, realized economic outcomes can be attributed, operational constraints can be expressed, and repeated feedback can be used to improve a decision policy. The category is intended for transaction-rich but frictional markets such as wholesale and retail inventory, microlending, used vehicles, freight, capacity allocation, thin digital assets, specialty insurance, and other businesses in which capital is repeatedly deployed into asset-like positions whose returns depend jointly on market conditions and operator performance.

The central proposal is a minimal portable interface called the **Menu-Sales-Description (MSD) contract**. A business menu is a decision-time table of candidate deals, sizes, features, observed profit where available, explicit missing outcomes where unavailable, and a historical-policy indicator such as historically_taken. A sales table is a keyed event tape of position realization: sales, repayments, closures, fees, returns, write-offs, liquidations, and the extraction of the base asset, normally cash. A business description states the accounting objective, operating workflow, costs, constraints, timing, market-impact assumptions, and legal or platform rules. At inference time, the operator supplies a current menu with the same schema and unknown profit; the system returns selected actions, sizes, predicted economics, uncertainty, and the option to take no action.

We state the **Computable-Market Thesis**: subject to support, attribution, feedback, modelable impact, and stability conditions, a sufficiently expressive and locally aligned nonparametric decision system can approach the best attainable use of the information contained in the MSD contract. The claim is information-relative, not a promise of perfect prediction, universal arbitrage, or guaranteed profit. Cross-market pretraining, table-semantic models, nonparametric outcome estimators, offline-policy correction, LLM-grounded simulation, and decision-focused portfolio optimization are presented as complementary components of such a system.

The paper integrates P34/PARML as an early implementation. In the executed slower-market-waves notebook, a conventional XGBoost classifier achieved 0.9266 ROC-AUC on a business-observed early-stop holdout and generated +$61.7 thousand in a naive observed-data evaluation, but lost $417.4 thousand when applied to the full 20-week future opportunity menu. On the reserved week-80 task menu, the executed P34 REST API run selected 115 trades, predicted +$2,404, and realized +$2,250 on $4,146 of deployed capital; the calibrated XGBoost baseline selected 177 trades and realized -$8,789 on $18,821. The underlying market is synthetic, and the one-menu API result is not evidence of live-market profitability or universal market computability.

> **Status and disclosure.** This paper is an integrated conceptual and technical working paper for industry and research review. It incorporates and extends the June 2026 P34/PARML technical paper. The reported market environment is synthetic. One result was obtained by an executed P34 REST API run on a withheld synthetic task menu; this is a live system invocation, not a live-market trial. The paper is not investment advice and does not disclose deployment-sensitive architecture, training compute, private data, private connectors, trading endpoints, or exact live control logic. Any operational use requires market-specific validation, independent review, shadow deployment, staged capital exposure, hard risk controls, and rollback procedures.

JEL classification: C45, C53, D81, G11, G14, L81, M11, M15
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1 Introduction

Many businesses perform activities that are economically equivalent to trading but are not organized as public exchanges. A wholesaler acquires inventory, takes demand and liquidation risk, and exits through sales. A microlender advances cash, services a receivable, and exits through repayment, recovery, or write-off. A dealership buys a vehicle, incurs financing and reconditioning costs, and exits through a sale. A freight broker, GPU-capacity allocator, insurance underwriter, customs-clearance guarantor, lead buyer, or thin-asset trader faces the same abstract problem: repeatedly allocate scarce capital or capacity among many candidate positions, then manage operational processes that transform uncertain future cash flows into realized yield.

These markets differ from the stylized frictionless market of elementary finance. The opportunity set is large; the profitable subset is often sparse; position size can change the sign of expected value; outcomes are observed mainly for actions taken by a biased historical policy; cash flows arrive with delay; the operator's own logistics, collections, marketing, financing, and execution determine asset value; and market impact or capacity constraints make return endogenous to scale. The task is therefore not merely to forecast a price. It is to construct a feasible portfolio from a high-cardinality, partially observed action set and to know when the correct portfolio is empty.

This paper proposes **computable market** as the name for this category when the market-operator pair can be exposed to a learning system through a repeatable data and decision loop. Computability here is not Turing computability and it is not a claim that every market can be solved. It is an operational-economic property: the extent to which an operator can represent feasible actions, reconstruct economic outcomes, encode constraints, and use feedback to reduce decision regret.

The proposed category develops the idea introduced in the P34/PARML technical paper: a model should learn directly against final economic outcomes rather than depend on a long chain of forecasts, dashboards, human rules, and approvals (Gree, 2026). **Profit-As-Regression Machine Learning (PARML)** denotes the broader model class. P34 denotes HyperC's current implementation, which combines market-regime detection, nonparametric profit estimation, false-positive correction, plausible-universe construction, drift hypotheses, and portfolio selection under partial observability.

The additional contribution of the present paper is to make the market interface explicit and minimal. The proposed **Menu-Sales-Description (MSD) contract** contains:

1. a historical **business menu** table representing candidate actions, sizes, features, economic outcomes where observed, and the prior policy signal `historically_taken`;
2. a **sales** table representing the keyed event history through which positions are opened, serviced, partially realized, closed, written off, and converted back into the base asset; and
3. a natural-language and machine-readable **business description** specifying accounting, operations, costs, constraints, timing, and market laws.

The current decision is configured by passing a new business menu with the same structure and unknown outcome. The output is an executable portfolio: actions, quantities, predicted profit or another economic target, uncertainty, policy diagnostics, and a no-trade decision where appropriate.

The strongest claim of this paper is deliberately narrower than universal profitability:

> **Computable-Market Thesis.** For a market-operator pair satisfying representability, outcome attribution, feedback, support, and modelable-impact conditions, there exists, with sufficient relevant data, engineering effort, and compute, a sequence of increasingly expressive, locally aligned decision systems whose policy regret approaches the best attainable policy measurable from the information contained in the MSD contract.

"Best attainable" means best relative to the supplied information, feasible actions, accounting objective, and governance envelope. Information that is absent, causally unidentified, stale, manipulated, or outside support cannot be recovered by compute alone. No-free-lunch results rule out a universally superior optimizer over arbitrary problem classes (Wolpert and Macready, 1997), and no-arbitrage theory rules out a general promise of riskless profit in well-specified financial markets (Delbaen and Schachermayer, 1998). Computable markets are valuable precisely because they impose structure: repeated menus, operator-specific telemetry, delayed but measurable outcomes, and exploitable regularities.

The paper makes six contributions. First, it gives an operator-relative definition and a practical computability test. Second, it describes the economic anatomy of the market category in professional terms. Third, it formalizes the MSD contract as a portable interface and distinguishes interface sufficiency from information sufficiency. Fourth, it describes cross-market pretraining and local alignment for statistically nonparametric decision systems. Fifth, it defines a constrained role for LLM agents in grounding business descriptions and constructing auditable simulation hypotheses. Sixth, it integrates P34/PARML and its existing benchmark evidence into the broader theory.

2 Terminology, Scope, and Relation to Prior Uses

2.1 A proposed operator-relative meaning

The phrase *computable market* has appeared previously in a different context. Mazzoli, Morini, and Terna (2019) use "A Computable Market Model" as the title of an agent-based macroeconomic simulation chapter. This paper does not claim that the two-word phrase has never appeared. It introduces and formalizes a distinct meaning centered on **operator-relative machine learnability and executable economic decision loops**. The unit of analysis is not a whole macroeconomy simulated from equations; it is a market-operator pair connected to repeated menus, event histories, and measurable business outcomes.

The proposed term is also distinct from several adjacent usages:

- A **digital market** may be electronically accessible but not learnable from operator data.
- An **algorithmic market** may use automated execution but expose no reliable counterfactual outcomes.
- A **liquid market** can be easy to execute in but informationally efficient and therefore difficult to profit from.
- A **computational market** often refers to markets for compute resources rather than to computability as a decision property.
- A **prediction market** aggregates beliefs through contracts; a computable market is a broader category that may include operational assets and non-security positions.
- A **simulation model** may reproduce stylized behavior without being linked to a live, economically accountable decision loop.

The adjective *computable* therefore describes a practical relationship among a market, an operator, a data contract, and a decision system. The same external market may be computable for one operator and not another because operators differ in access, telemetry, accounting quality, execution capability, risk constraints, and scale.

2.2 Market, operator, asset, and action

Let a market expose at decision time t a set of candidate state-action configurations

$$A_t = \{a_{t1}, \dots, a_{tn_t}\}.$$

An action is not merely an asset identity. It can include the asset, counterparty, quantity, price, timing, channel, financing plan, execution method, exit rule, and operational treatment. Two quantities of the same asset are different actions when liquidity, capacity, or cost is nonlinear. A portfolio p_t is a feasible subset or weighted combination of actions subject to capital, concentration, mutual-exclusion, inventory, compliance, and operational constraints.

The **operator** o is the business or investor whose capabilities transform actions into outcomes. Realized economic outcome is therefore written

$$Y_{t+h}(p_t; o, \omega),$$

where h is the realization horizon and ω denotes market and operational uncertainty. The operator argument matters. The same inventory lot, loan application, vehicle, shipment, or token position can have different values for two businesses because their cost structures, conversion rates, collection capabilities, financing costs, and exit channels differ.

2.3 Computability is graded, not binary

A practical market can be more or less computable. We define a market as (ϵ, δ, T)**-computable for operator** o **under information contract** I when there exists a feasible policy $\hat{\pi}$ that, over horizon T, achieves regret no greater than ϵ relative to the best policy measurable from I, with probability at least $1-\delta$, under the stated deployment distribution and controls. This definition is aspirational in markets where the counterfactual optimum is unobservable; it becomes testable through synthetic generators, randomized exploration, shadow deployments, and bounded live experiments.

The definition emphasizes three points. First, computability is relative to the operator and information contract. Second, it is relative to a performance tolerance and horizon. Third, it concerns decisions, not perfect state reconstruction. A model can be economically useful even when it is uncertain about many individual outcomes if it reliably avoids negative portfolios and captures a sufficient fraction of the positive opportunity set.

2.4 Candidate conditions for a computable market

Table 1 gives a practical screening test. Failure of one condition does not always make the market permanently non-computable; it may indicate that the first investment should be instrumentation rather than modeling.

Table 1. Candidate conditions for operator-relative computability.

Condition	Professional interpretation	Diagnostic question
Representable actions	Feasible state-action configurations can be materialized as rows or structured sets.	Can the business enumerate what it could actually do at a decision time?
Attributable outcomes	Cash flows, costs, defaults, returns, closures, and write-offs can be linked to decisions.	Can realized economics be reconstructed without arbitrary allocation?
Historical policy signal	The logging or prior business policy is observable at least through selected actions.	Do we know which rows were available and which were taken?
Repeated feedback	Comparable decisions recur often enough to support evaluation and adaptation.	Are there enough cycles to distinguish luck from policy quality?
Expressible constraints	Budget, capacity, compliance, mutual exclusion, and concentration rules can be encoded.	Can a proposed portfolio be checked for executability?
Support or safe extrapolation	Deployment actions have historical support, safe grounding, or bounded simulation evidence.	Is the model being asked to act far outside observed scale or behavior?
Limited or modeled impact	Taking an action has bounded, observable, or simulable effects on prices and future menus.	Does execution destroy the validity of the remaining menu in an unobserved way?
Testability and reversibility	Holdout, forward, shadow, or staged live evaluation is possible with controlled downside.	Can the policy be falsified and rolled back?
Stable economic accounting	The target is defined consistently in units meaningful to the operator.	Does "profit" include financing, labor, returns, taxes, and liquidation on the same basis?
Governance compatibility	Legal, contractual, platform, and risk limits can constrain the action space.	Can automation be bounded before it reaches capital or customers?

3 Economic Anatomy of Computable Markets

3.1 Transaction-rich but opportunity-sparse

The first defining property is a large opportunity set with a comparatively small positive-risk-adjusted subset. A distributor can see thousands of stock-keeping-unit, supplier, price, quantity, and timing combinations. A lender can see thousands of applicant-amount-term combinations. A dealer can evaluate many vehicles at many bids and reconditioning levels. Most configurations may be unattractive after all costs, risks, and capacity effects are included.

Let gross economic value of action a for operator o be

$$G(a,o) = \text{revenue} - \text{acquisition cost}.$$

The decision-relevant value is closer to

$$V(a,o,q) = G(a,o,q) - C_{\text{execution}} - C_{\text{operations}} - C_{\text{funding}} - E[L] - C_{\text{exit}} - \rho(a,o,q),$$

where q is size, $E[L]$ is expected loss, and ρ is a risk, liquidity, or capital charge. A market can contain abundant transactions while the set $\{a : V(a,o,q) > 0\}$ is sparse. The scarce factor is not raw deal flow but reliable screening capacity.

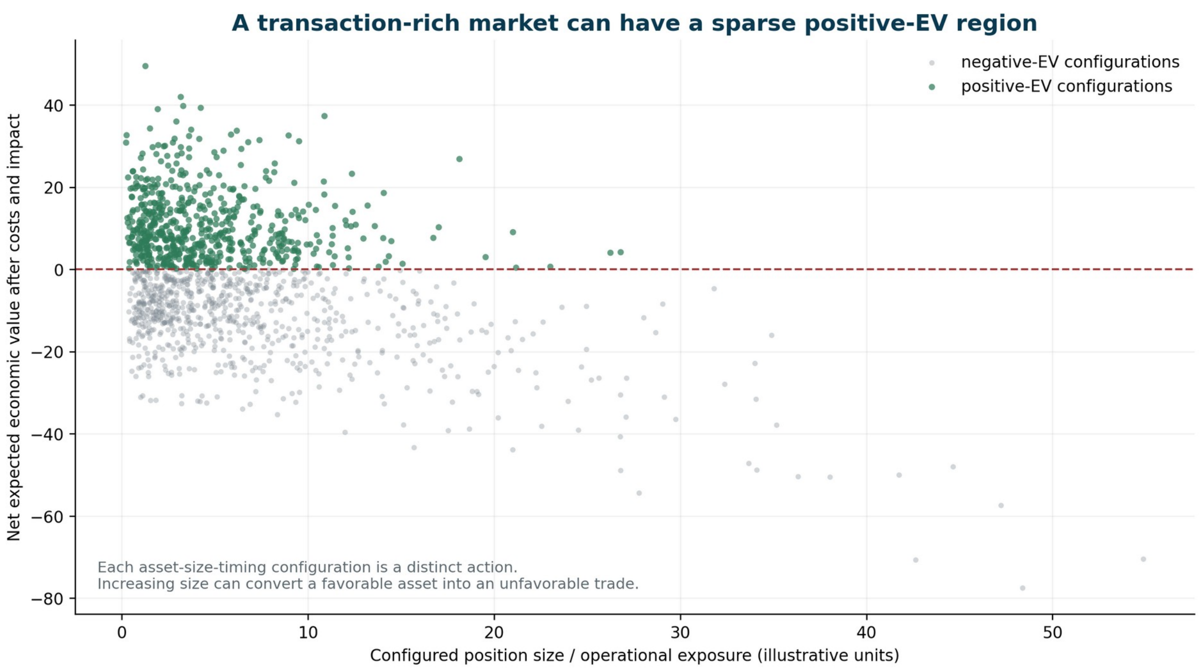

Figure 1. A transaction-rich market can contain a sparse positive-EV region once size, operating cost, and impact are treated as part of the action. The distribution is illustrative.

The professional description is **a high-cardinality opportunity set with sparse positive expected value**. "Long tail" is a useful commercial metaphor, but the statistical distribution need not be heavy-tailed. The precise claim is that profitable configurations are dispersed across a large heterogeneous cross-section and cannot be captured by one threshold or one obvious mode.

3.2 State-action configurations rather than standalone assets

A listed ticker at a quoted price is not yet a complete decision. "Buy the asset" hides quantity, timing, execution, financing, hold period, exit policy, and interaction with existing exposure. In a thin order book, buying \$5,000 and buying \$500,000 are different economic objects. In wholesale, buying 100 units and 20,000 units can reverse expected profit through inventory aging and liquidation. In lending, increasing loan size changes both exposure and borrower behavior. In insurance, aggregating correlated policies changes tail risk.

The relevant object is therefore a **state-contingent action configuration**. This makes the action space combinatorial and explains why thousands of nominal deals can become millions of economically distinct deal-size-timing choices.

3.3 Operator-dependent and operationally mediated returns

In conventional asset-pricing exposition, the asset's return is often treated as external to the investor. In computable operational markets, the operator participates in producing the return. Expected value depends on underwriting, pricing, advertising, fulfillment, service quality, collections, reconditioning, tax handling, inventory placement, and exit capability. We call this **operationally mediated return** and the corresponding valuation **operator-dependent asset value**.

Formally, two operators o_1 and o_2 can rationally assign different values to the same action:

$$V(a, o_1, q) \neq V(a, o_2, q).$$

This difference is not necessarily disagreement about the external market. It can be comparative advantage. One operator may have lower shipping cost, better collections, a stronger lead-conversion channel, cheaper capital, or more liquid exit access. A computable-market model must therefore learn both market behavior and the operator's production function.

3.4 Nonlinear sizing, strategy capacity, and market impact

Volume usually increases absolute profit only up to the point at which marginal opportunity quality, execution cost, or operational capacity deteriorates. The professional terms are **capacity-constrained returns**, **nonlinear transaction costs**, **market impact**, and **declining marginal alpha**. Market microstructure has long recognized that depth and price impact make quantity part of valuation (Kyle, 1985), while optimal-execution models jointly consider volatility risk and permanent and temporary impact (Almgren and Chriss, 2001).

In an operational market, the same logic appears through different channels: warehouse limits, working-capital covenants, demand saturation, ad-auction competition, service capacity, collection staffing, geographic concentration, or liquidation discounts. A model that predicts unit economics but treats quantity as an afterthought can select a portfolio that is profitable in small samples and destructive at deployment scale.

3.5 Partial observability and policy-selected labels

Businesses rarely observe the outcome of every action they could have taken. Reliable outcomes are concentrated among actions selected by a historical policy of humans, rules, or prior models. The label process is therefore not random. This is a selection problem in the sense of Heckman (1979), a logged-bandit problem in the sense of counterfactual risk minimization (Swaminathan and Joachims, 2015), and an out-of-distribution problem in offline reinforcement learning (Fujimoto, Meger, and Precup, 2019; Kumar et al., 2020).

The observed dataset can be optimistic for two reasons. First, the business attempted to accept good opportunities. Second, only accepted opportunities reveal realized outcomes. A conventional regressor can fit this selected sample accurately and still over-enter when exposed to a wider deployment menu. The economically important error is the **false positive**: an action predicted to be profitable that realizes a loss after hidden downside, drift, or execution failure.

The `historically_taken` field is therefore not an outcome label. It is a behavior-policy signal. It helps estimate what the prior business considered acceptable, where support exists, and how operational risk appetite shaped observation. The model must be able to learn from this signal without simply copying the old policy.

3.6 Delayed, path-dependent realization

Economic outcome is often not available at the decision time and may not be reducible to one terminal price. Inventory sells in pieces; a loan produces installments and collections; a vehicle accumulates financing and repair costs before sale; a customs policy remains open until clearance or claim resolution; a thin asset exits through a sequence of fills. The final label is a function of a path of events:

$$Y_i = \sum_{e \in E_i} cashflow(e) - allocated\,costs(e),$$

where E_i is the event history linked to position i.

This is why the proposed sales table is an **event tape**, not merely an aggregate revenue table. It preserves timing, partial realization, fees, returns, write-offs, and closure state. Historical simulation becomes difficult when future liquidity, customer behavior, or operational timing must be modeled rather than looked up.

3.7 High decision frequency and low error tolerance

Computable markets often contain thousands to millions of annual allocation decisions. They may be low-margin, so small calibration errors compound over volume; or high-downside, so a small number of false positives dominates performance. The appropriate objective is not necessarily lowest row-level mean squared error. It can be portfolio profit, downside-adjusted return, false-positive rate, capital efficiency, or a constrained utility function. Decision-focused learning explicitly recognizes that prediction quality should be evaluated through the downstream optimization problem (Elmachtoub and Grigas, 2022).

3.8 Economies of scale in information processing

In these markets, more volume can produce more profit only if the operator can score more of the opportunity set without lowering decision quality. Automation creates **economies of scale in screening, simulation, and portfolio construction**. The value of compute is not that it creates economic regularities from nothing; it allows the system to evaluate more configurations, preserve more conditional structure, run more plausible worlds, and update more frequently than a human organization can.

The Black-Scholes-Merton framework is an early emblematic example of computation changing market practice: a tractable model made a large space of contingent claims more consistently comparable (Black and Scholes, 1973; Merton, 1973). Computable-market systems seek a broader, empirical analogue for operational assets, without assuming one closed-form law.

4 Formal Framework and the Computable-Market Thesis

4.1 The information set

Let B_o denote the operator's business description; $M_{\le t}$ the history of menus; $S_{\le t}$ the sales/event tape; and M_t^{cur} the current menu. Define the decision information set

$$I_{o,t} = \sigma\left(B_o, M_{\le t}, S_{\le t}, M_t^{cur}\right).$$

The feasible policy class Π_o maps this information to portfolios satisfying operator constraints. Let $U_o(Y, p)$ be the configured utility, normally a function of profit, downside, capital use, concentration, liquidity, and policy penalties. The information-relative optimal policy is

$$\pi_o^{\text{¿}}(I_{o,t}) \in \arg\max_{\pi \in \Pi_o} E\left[U_o\left(Y_{t+h}\left(\pi(I_{o,t})\right), \pi(I_{o,t})\right) \mid I_{o,t}\right].$$

The objective does not require the model to know a metaphysically complete market state. It requires the model to use the available information no worse than other feasible policies, subject to estimation and optimization error.

4.2 Interface sufficiency versus information sufficiency

The paper's minimality claim is an **interface claim**, not an assertion that any two tables contain enough signal. If the business description, menu history, and event tape generate the decision-relevant information set, then any Bayesoptimal policy can be expressed as a measurable function of that triple. Nothing conceptually requires a fourth interface. Additional telemetry can be added as menu columns, event attributes, or referenced context in the description.

Proposition 1 (MSD interface sufficiency). Suppose the conditional distribution of future economic outcomes for every feasible current action depends on the operator's available records only through $I_{o,t}$. Then an information-relative optimal policy is $I_{o,t}$-measurable and can, in principle, be implemented from the MSD contract.

Interpretation. The proposition is almost definitional, but useful. It identifies a stable systems boundary. A menu supplies the cross-sectional choice set; a sales tape supplies longitudinal realization; a description supplies semantics, accounting, and constraints. Whether the supplied records are actually rich enough is an empirical question.

4.3 Best use of information

The phrase "best use of information" can be stated through regret. Let

$$V_T(\pi) = E\left[\sum_{t=1}^{T} U_o(Y_{t+h}, p_t)\right)$$

and define information-relative regret

$$R_T(\hat{\pi}) = V_T(\pi_o^{¿}) - V_T(\hat{\pi}).$$

A system uses information better when it reduces this regret under the same feasible action set, accounting basis, and governance envelope. More validated information cannot reduce the theoretical optimum because the policy can ignore it; in Blackwell's decision-theoretic sense, a more informative experiment weakly expands attainable decision performance (Blackwell, 1951). In finite samples, however, additional features can worsen estimation through noise, leakage, or overfitting. Computable-market engineering must therefore combine broad ingestion with pruning, stability tests, and portfolio-level validation.

4.4 A static consistency result

A stylized result clarifies what "enough engineering and compute" can and cannot mean.

Proposition 2 (Static information-relative consistency). Consider a finite current action set and bounded utility. Assume: (i) every action considered at deployment has adequate support or a valid identification strategy; (ii) realized labels are consistently reconstructed; (iii) the conditional expected utility of each action is estimable by a universally consistent nonparametric procedure; and (iv) optimization error converges to zero. Then the plug-in policy that chooses the feasible portfolio with maximum estimated conditional utility has regret converging to zero relative to the information-optimal policy.

Proof sketch. Universal consistency yields convergence of estimated conditional utilities to their true conditional values under the information set (Stone, 1977). Uniform convergence over the finite feasible action set and vanishing optimization error imply that the value of the selected action converges to the maximum conditional value. The gap between the selected and optimal value therefore converges to zero. The result extends to controlled portfolio classes under standard uniform convergence and continuity conditions.

This proposition supports the existence claim in a restricted setting. It does not solve dynamic endogeneity, nonstationarity, interference among actions, strategic adversaries, or unsupported extrapolation. Those require market-specific structure and are the reason the paper defines computability through explicit conditions rather than through compute alone.

4.5 The dynamic computability hypothesis

The practical claim is a research hypothesis about a sequence of systems, not one fixed model:

$$F_1 \subset F_2 \subset \cdots,$$

where capacity can grow through retrieval, trees, kernels, ensembles, larger context, more simulated universes, richer meta-learning, or differentiable components. As data quality, market coverage, simulator fidelity, and compute increase, the system should reduce four error terms:

$$R \lesssim E_{\text{representation}} + E_{\text{estimation}} + E_{\text{simulation}} + E_{\text{optimization}}.$$

Engineering effort matters because each term is partly a systems problem. Compute can expand candidate portfolios and hypotheses; it cannot repair undefined accounting, unlinked identities, prohibited experimentation, or unobserved market impact without new information.

4.6 What nonparametric means here

A pretrained neural network has a finite parameter count and is therefore parametric in a literal engineering sense. This paper uses **nonparametric decision system** in the statistical and operational sense: the market response is not restricted to a fixed low-dimensional equation chosen ex ante, and effective capacity can grow with data, retrieval, ensembles, tree depth, kernels, simulation budgets, and meta-learning. The system may be hybrid or semiparametric. The important distinction is from assuming that all markets share one closed-form pricing function with a fixed small parameter vector.

4.7 Boundary conditions

A market may be non-computable, or only weakly computable, when:

- outcomes cannot be attributed to decisions;
- feedback arrives too late relative to regime change;
- deployment actions lie outside all historical or simulated support;
- one action changes the entire future opportunity set in an unobserved way;
- accounting omits material costs or allocates them inconsistently;
- strategic participants manipulate the telemetry;
- the operator cannot encode or enforce legal and operational constraints;
- the target is dominated by irreducible shocks rather than stable conditional structure; or
- safe evaluation is impossible.

In such cases, the first deliverable is better instrumentation, not a more confident model.

5 The Menu-Sales-Description Data Contract

5.1 Why a minimal contract matters

A general cross-market model cannot require every business to rebuild its data warehouse around one industry ontology. It needs an interface that is small enough for ordinary business IT teams to produce, but expressive enough to preserve decision structure and economic realization. The MSD contract separates three kinds of information that are frequently mixed in operational systems:

- **what could have been done** at each decision point;
- **what subsequently happened** to positions and cash flows; and
- **what the business means** by actions, costs, closure, constraints, and success.

The separation is important. A menu without an event tape becomes a static supervised-learning table that often hides delayed realization. An event tape without a menu records history but not the alternatives that were rejected. A business description without structured records supports narrative analysis but not reproducible portfolio learning. Together, the three objects form a compact market interface.

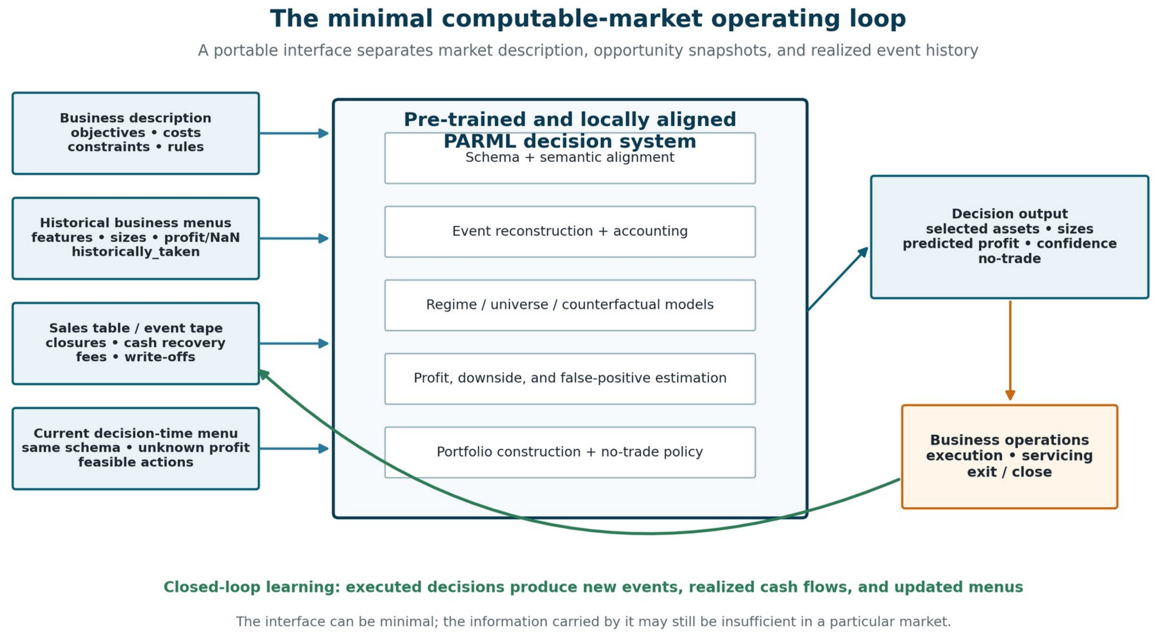

Figure 2. The minimal computable-market loop. Historical menus, the sales event tape, the current menu, and the business description are aligned by a pre-trained PARML decision system; executed actions create the next round of observed events.

5.2 The business menu as a market snapshot

A **business menu** is a table representation of feasible actions available to the operator at a decision time. Each row should describe a complete action configuration at the granularity relevant to choice. Rows can represent direct instruments, quantity alternatives, timing choices, suppliers, channels, bids, sub-portfolios, or already assembled portfolios. A menu can be daily, vendor-specific, channel-specific, event-specific, or a rolling virtual window.

The historical menu serves two roles. It is a cross-sectional **market snapshot**, and it records the choice set from which the historical policy acted. The current menu uses the same semantics but normally has no realized outcome.

A canonical minimal schema is shown in Table 2. Column names can vary if semantic mappings are supplied, but preserving these roles makes transfer and auditing easier.

Table 2. Canonical business-menu fields.

Field	Status	Meaning
menu_id or bag_id	Required	Groups mutually related or jointly constrained actions available at one decision point.
decision_time or date	Required	Time at which the action was available and information was known.
asset_id or ticker	Required	Stable identity of the position, instrument, product, lead, loan, vehicle, shipment, or other asset-like object.
action_id	Recommended	Unique identity for the fully configured action; can distinguish sizes, channels, counterparties, and timing.
size or qty	Required	Position size, units, principal, coverage, bid quantity, capacity, or another exposure measure.

Field	Status	Meaning
profit	Required for a unified interface; may be missing	Realized economic outcome where known; NaN where unobserved, open, or counterfactual. The accounting basis must be defined.
historically_taken	Strongly recommended	Indicator or probability showing whether the historical policy executed the action. It is a policy signal, not a profitability label.
menu_available	Recommended	Indicates whether the action was actually available to this operator, rather than observed elsewhere or synthetically generated.
feature columns	Required in aggregate	Prices, costs, demand, counterparty, quality, timing, risk, inventory, external signals, and other decision-time telemetry.
constraint columns	Recommended	Capital use, mutual-exclusion group, capacity consumption, compliance flags, or execution limits.
label provenance	Recommended	Observed, safely grounded, allocated, simulated, imputed, or pending.

A row must contain only information available at decision time, except for clearly separated outcome and provenance fields. Leakage checks are not an optional refinement: post-decision inventory, realized sales velocity, recovery status, or eventual market prices can create convincing but unusable backtests.

5.3 Missing outcomes are counterfactuals, not zeros

The profit field is populated where the outcome is observed or safely reconstructed. It is NaN where the action was not taken, remains open, or cannot yet be attributed. Missingness is semantically meaningful. It indicates a counterfactual or unresolved outcome, not zero profit.

A business can sometimes onboard without explicitly materializing every missing row. For example, the business description and historical event systems may allow a connector to reconstruct historical candidate menus or generate quantity alternatives. In that sense, explicit NaN cells are not logically necessary for every unchosen action. Nevertheless, **outcome observability must remain explicit** through a missing-value field, observation flag, or provenance code. A model-generated pseudo-label must never become indistinguishable from a realized cash-flow label.

Safe grounding can expand observed labels. If a batch of N units was executed and the first $m < N$ units can be identified with the same acquisition cost and realized sales sequence, then the smaller quantity may have a defensible grounded outcome. Similar logic may apply to partial principal amounts, capacity blocks, or nested coverage levels. Grounding beyond observed scale is more dangerous and should normally be treated as simulation, not fact.

5.4 The sales table as a generalized event tape

The name **sales table** is retained because it is intuitive in commerce, but the semantic object is broader: it is an append-only or replayable **position-realization event tape**. Each event records a change in exposure, cash, liability, cost, status, or closure state. The tape allows the system to reconstruct profit paths, holding periods, partial exits, working-capital use, and delayed outcomes.

Table 3. Canonical sales/event-tape fields.

Field	Status	Meaning
event_id	Required	Unique event identity for deduplication and audit.

Field	Status	Meaning
asset_id or position_id	Required	Links the event to the menu identity or opened position.
action_id or origin_menu_id	Recommended	Links realization to the exact decision when possible.
event_time	Required	Timestamp or business date of the event.
event_type	Required	Open, purchase, disbursement, sale, repayment, fee, return, default, recovery, liquidation, write-off, close, or market-specific equivalent.
quantity_delta	Recommended	Change in units, principal, coverage, or outstanding exposure.
cash_flow_base	Required where economic	Signed cash flow in the configured base asset; positive normally means cash extracted, negative means capital deployed or cost incurred.
currency or base_asset	Required when multiple	Unit of account and conversion basis.
fee, tax, operating_cost	Recommended	Separately recorded costs where available; can also be encoded as event types.
status_after	Recommended	Open, partially realized, closed, defaulted, written off, cancelled, or another defined state.
source_system and provenance	Recommended	Origin, correction history, and confidence for reconciliation.

The event tape should support partial closure. A position may produce several sales, returns, reimbursements, payments, or recoveries before it is economically complete. "Closed" must be defined by the business description: zero inventory, maturity, claims resolution, collections cutoff, legal release, or another rule.

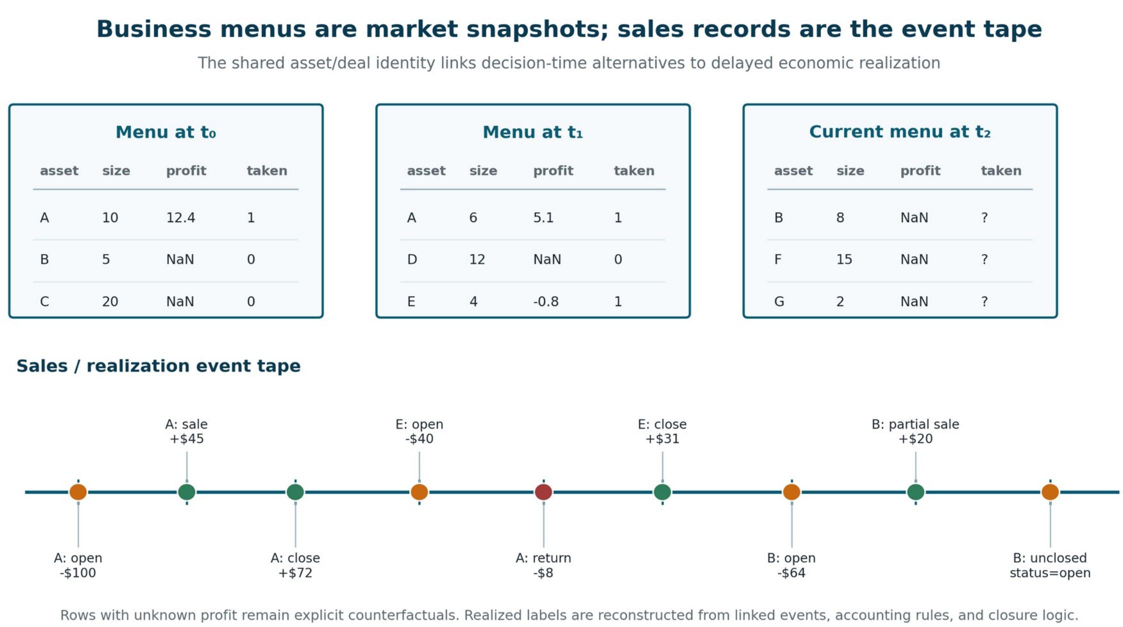

Figure 3. Menus provide cross-sectional snapshots of available actions; the sales table provides the longitudinal event tape. Shared identities and accounting rules reconstruct realized labels and distinguish unobserved counterfactuals.

5.5 Reconstructing realized profit

For position i, a generic realized-profit label can be reconstructed as

$$profit_i = \sum_{e \in E_i} cash_{¿}(e) - \sum_{e \in E_i} allocated_{¿}(e) - capital\,charge_i\,,$$

subject to the accounting constitution of the business. The formula can include inventory mark-downs, accrued interest, labor allocation, platform fees, taxes, expected residual liability, or a terminal valuation for open positions. A model cannot infer the economically correct target if the organization itself has not defined it.

The tape also allows alternative labels without changing the interface: contribution margin, internal rate of return, recovery rate, liquidation value, days-to-cash, probability of full repayment, downside loss, or utility-adjusted profit. PARML means that the model is trained toward the final business objective rather than one predetermined accounting metric.

5.6 The business description as an operational constitution

The **business description** is a versioned description of the market and operator. It can begin as natural language, diagrams, contracts, fee schedules, and standard operating procedures, but should be compiled into machine-readable rules wherever possible. It supplies the semantics that column names and events cannot carry reliably across markets.

At minimum it should define the elements in Table 4.

Table 4. Recommended content of the business description.

Domain	Required content
Economic objective	Profit or utility definition, unit of account, timing convention, capital charge, treatment of open positions, and risk penalties.
Asset and position ontology	What constitutes an asset, action, position, portfolio, open event, partial realization, and final closure.
Workflow	Acquisition, execution, servicing, fulfillment, collection, claim, liquidation, and exception processes.
Costs and revenues	Price formation, fees, shipping, financing, labor, advertising, returns, taxes, claims, write-offs, and shared-cost allocation.
Constraints	Budget, leverage, inventory, concentration, capacity, compliance, geography, customer, platform, and counterparty limits.
Timing and delay	Lead times, settlement, sales cycles, repayment schedules, holding periods, expiration, and feedback latency.
Market impact and interference	How quantity, execution, competition, or one action changes prices, availability, or other actions.
Historical policy	How humans or systems selected deals, including exploration, thresholds, approvals, and known policy changes.
Data semantics	Units, keys, missingness, data provenance, known leakage risks, and source-system reliability.
Governance	Allowed markets and actions, escalation rules, audit requirements, overrides, kill switches, and deployment stages.

The description should be treated as code-adjacent infrastructure: versioned, reviewable, testable, and linked to the data interval in which it applied. A fee schedule or operating rule can change; using the current description to simulate old menus can create retrospective leakage.

5.7 The current decision-time menu and output contract

At inference time the operator sends a current menu with the same structural roles as the historical menu. Fields such as asset_id, decision_time, size, feature columns, availability, and constraints are populated. The realized profit field is unknown and should remain missing.

For ergonomic compatibility, the model can return the same rows with predicted values added. For auditability, it is preferable to keep observed and predicted fields separate:

Table 5. Recommended decision-output fields.

Field	Meaning
selected	Whether the action belongs to the executable portfolio.
selected_size	Final size after portfolio constraints and mutual exclusion.
predicted_profit	Conditional expected economic outcome under the configured accounting basis.
prediction_interval or quantiles	Uncertainty range, including downside estimates.
probability_negative	Estimated probability of a negative realized outcome.
support_score	Proximity to observed or safely simulated policy support.
regime_id or state	Market/business regime used for the decision.
policy_reason_code	No-trade, capital limit, dominated action, false-positive correction, compliance, or other machine-auditable reason.
simulation_provenance	Whether and how LLM-grounded or synthetic hypotheses affected the estimate.
decision_version	Model, data, description, and rule versions needed for replay.

The no-trade action should always be representable unless a business process legally or operationally requires allocation. In low-margin markets, declining all current opportunities can be the highest-value portfolio.

5.8 historically_taken as behavior-policy and alignment telemetry

The historically_taken column records what the prior business policy did. In its simplest form it is binary. Richer versions can include selected quantity, approval stage, rejected reason, policy score, or estimated selection probability. It serves four purposes:

1. **support diagnosis:** identifying regions where the model has real outcome evidence;
2. **selection-bias correction:** estimating propensities or conservative penalties for unobserved actions;
3. **operator alignment:** revealing historical risk appetite, turnover preference, capital discipline, and operational capacity; and
4. **policy comparison:** measuring whether a new model changes the business in economically intended ways.

Historical behavior is not automatically desirable. It can encode obsolete constraints, hidden discrimination, inconsistent human judgment, or systematic underinvestment. The field should therefore be treated as a prior and diagnostic, not as the target policy. A complete theory of policy inheritance and risk-appetite alignment is outside the scope of this paper, but the field belongs in the minimal contract because it makes the logged-data problem explicit.

5.9 Identity, units, and temporal integrity

Most failed market models are not defeated first by architecture; they are defeated by identity and time. Menu rows and tape events must share stable keys. Units must be explicit. Currency conversion must use decision-appropriate rates. Features must be timestamped by when they became knowable. Corrections and late-arriving events must be replayable. Corporate actions, product merges, refinancings, split shipments, partial returns, and reassigned leads require identity lineage.

A high-quality MSD implementation therefore includes schema validation, unit tests, accounting reconciliation, duplicate detection, open-position aging, late-event policy, and date-aware data splits. These engineering controls are part of market computability, not merely data hygiene.

6 Cross-Market Pretraining and Local Alignment

6.1 The case for a pre-trained market decision model

A market-specific model can learn one dataset from scratch, but many relevant structures recur across industries: size-dependent profit, delayed realization, censored outcomes, policy-selected labels, survival and default, market impact, capital constraints, portfolio concentration, regime change, and the value of waiting. A cross-market model can pretrain on these structural regularities while leaving local semantics and economics to alignment.

Recent tabular foundation-model research makes the direction technically plausible. TabPFN learns a tabular prediction algorithm across millions of synthetic datasets and performs in-context prediction on new tables (Hollmann et al., 2025). CARTE represents rows as context-aware graphs and supports transfer across tables with unmatched schemas using column and entry semantics (Kim, Grinsztajn, and Varoquaux, 2024). XTFormer explicitly studies cross-table pretraining over heterogeneous feature-target mappings (Chen et al., 2024). These systems do not solve computable-market decisioning, but they demonstrate that reusable priors and semantic transfer across heterogeneous tables are no longer purely speculative.

A computable-market foundation model differs in four ways. It must ingest sets of candidate actions rather than isolated independent rows; condition on longitudinal event histories; correct logged-policy bias; and optimize portfolio economics under constraints. Its output is a decision, not merely a prediction.

6.2 Pre-alignment and pretraining across markets

We distinguish **pretraining** from **alignment**.

Pretraining exposes the system to many historical and synthetic market contracts. The model learns reusable priors about accounting patterns, delay, size curves, censoring, regimes, selection mechanisms, uncertainty, and portfolio effects. Synthetic markets are useful because they reveal complete counterfactual outcomes and can deliberately vary hidden bias, liquidity, and drift.

Pre-alignment teaches the model how to interpret the contract: which fields are identities, dates, sizes, outcomes, events, constraints, and prior-policy signals; how units and signs work; and how a current menu differs from a historical realized label. Semantic encoders use column names, descriptions, data types, distributional fingerprints, and the business description to map arbitrary enterprise schemas into common roles.

Local alignment adapts reusable priors to the operator. It learns the business's actual cost surface, access, efficiency, risk appetite, execution, and historical selection policy from the local MSD contract. Cross-market priors should regularize low-data regions, not override strong local evidence.

6.3 A multi-lifted architecture

Figure 4 shows a conceptual architecture. The term **multi-lifted** means that representations are progressively lifted from raw tables to economic states, from states to plausible market universes, and from universes to executable policies. Compute and engineering can be added at clear interfaces.

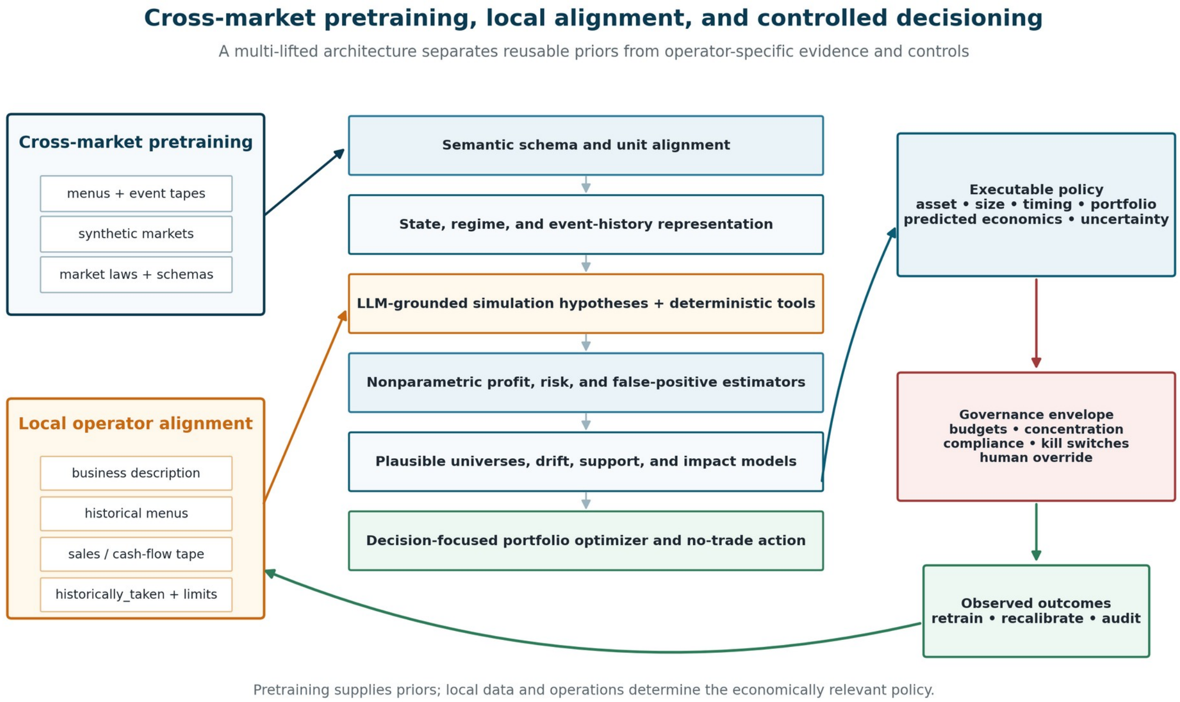

Figure 4. Conceptual cross-market architecture. Pretraining provides reusable priors; local operator evidence and controls determine the economically relevant decision policy.

The major layers are:

1. **Semantic schema and unit alignment.** Resolve identities, timestamps, quantities, currencies, target semantics, missingness, and constraints. Detect incompatible scales and leakage.
2. **State, regime, and event-history representation.** Encode current menus jointly with sales histories, open positions, market context, and operator state.
3. **Simulation-grounding layer.** Convert the business description into auditable rules, feature transformations, and candidate counterfactual hypotheses. Use deterministic tools for arithmetic and event replay.
4. **Nonparametric outcome and risk estimation.** Estimate profit, downside, closure time, liquidity, false-positive risk, and other labels without imposing one fixed market equation.
5. **Plausible universes and drift models.** Maintain multiple interpretations of unobserved actions, historical selection, regime, and future distribution; remove dominated hypotheses.
6. **Decision-focused portfolio optimization.** Select actions and sizes under capital, mutual-exclusion, concentration, capacity, and compliance constraints; include no-trade.
7. **Governance and feedback.** Enforce limits, log reasons, reconcile execution, observe outcomes, and recalibrate or roll back.

6.4 The role of broad feature ingestion

A general market model should be able to accept broad telemetry because new signals often arrive before an organization knows whether they matter. P34 calls this engineering principle **careless feature engineering**: rapid ingestion combined with stage-level pruning, stability testing, leakage review, and portfolio validation. The phrase does not license careless data management. It means the architecture should lower the cost of trying signals while retaining the ability to reject them.

Validated signals can include naive profit estimates, demand forecasts, risk flags, competitor measures, external prices, weather, market indicators, counterparty scores, and LLM-extracted attributes. A weak signal can still be useful as a reference for correction. A seemingly strong signal should be discarded if it fails date-aware forward tests or changes portfolio behavior without improving realized economics.

6.5 Nonparametric local heads and reusable representations

A practical implementation need not choose between foundation models and tree-based methods. A semantic encoder can produce reusable representations while local boosted trees, forests, kernels, nearest-neighbor estimators, survival models, or calibrated neural heads learn operator-specific relationships. The effective system is modular. P34 currently uses boosted and other nonparametric components over heavily materialized matrices in several blocks because they are debuggable and efficient for iteration; the longer-term architecture can make more of the signal flow differentiable.

6.6 The no-trade policy and conservative extrapolation

Cross-market pretraining increases the temptation to act outside local support. The correct response is not blanket rejection of transfer but explicit support telemetry. The model should know whether a recommendation is grounded in local observations, safe quantity nesting, closely related markets, simulator assumptions, or distant latent analogy. Conservative offline-learning methods address the risk of value overestimation outside the logged distribution (Kumar et al., 2020). In computable markets, support penalties, uncertainty, and the no-trade action are first-class outputs.

7 LLM-Grounded Operational Simulation

7.1 Why the business description needs an agentic parser

Operational markets are governed by language: fee schedules, contracts, platform rules, standard operating procedures, exceptions, underwriting policies, shipping terms, claim definitions, and human explanations of how work actually happens. Requiring an enterprise to hand-encode every rule before model onboarding makes cross-market deployment prohibitively slow. An LLM agent can reduce this cost by translating the business description into structured hypotheses, schema mappings, validators, and executable simulation plans.

The agent's job is not to declare profit. Its job is to turn narrative knowledge into **testable computational objects**. ReAct-style systems demonstrate a general pattern in which language models interleave reasoning with calls to external tools and environments (Yao et al., 2023). Research on generative agents and simulated economic agents shows that LLMs can produce behaviorally useful simulations, while also emphasizing that internal world models can be inconsistent, prompt-sensitive, and unsuitable as unvalidated causal evidence (Park et al., 2023; Horton, Filippas, and Manning, 2026).

7.2 A constrained simulation-grounding workflow

A safe workflow separates interpretation from arithmetic and validation:

1. **Parse.** Extract asset ontology, events, units, accounting rules, timing, constraints, and claimed causal mechanisms from the description.
2. **Map.** Propose mappings from enterprise columns and event types to canonical MSD roles.
3. **Compile.** Generate deterministic code, SQL, state machines, or rules for label reconstruction and constraint checking.
4. **Hypothesize.** Propose a bounded family of counterfactual mechanisms for missing outcomes, scaling, delays, and market impact.
5. **Simulate with tools.** Execute numerical simulation outside the language model using versioned inputs and reproducible random seeds.
6. **Back-check.** Compare simulated paths with observed menus, event tapes, holdouts, and known accounting identities.
7. **Calibrate and weight.** Treat each simulator as one universe with uncertainty and evidence weight; discard dominated or contradicted universes.
8. **Audit.** Retain the description version, extracted rules, generated code, tests, and model influence for each decision.

This design makes the LLM a **hypothesis and interface generator**, not a trusted numerical oracle.

7.3 Making grounded tables easier to learn from

Business descriptions improve tabular learning in three ways. First, they explain column semantics that raw numbers cannot reveal. A value of 30 might mean days, dollars, percent, miles, or units. Second, they allow the system to derive economically meaningful features such as age, outstanding principal, effective margin, exposure concentration, or closure state. Third, they support candidate pseudo-labels and scenario variables for unobserved actions.

Every derived or simulated field should carry provenance. A useful taxonomy is:

- `observed`: directly recorded and reconciled;
- `safely_grounded`: implied by a verified nested or accounting relation;
- `allocated`: derived through a documented shared-cost rule;
- `simulated`: generated by a stated mechanism;
- `imputed`: statistically filled without causal claim;
- `pending`: outcome not yet mature;
- `unobserved`: no label or defensible approximation.

A model can learn jointly from these sources by weighting them differently or treating provenance as a feature. What it must not do is collapse them into one indistinguishable target column.

7.4 Simulation as a bridge, not a substitute for telemetry

An LLM-grounded simulator is most valuable early in market onboarding and in sparse regions. As real execution produces new event-tape evidence, the model should relearn the market and operator dynamics and reduce dependence on speculative universes that fail calibration. The long-run authority is realized economic telemetry under controlled deployment, not linguistic plausibility.

Simulation can also test computability itself. If no plausible simulator family can reproduce basic observed invariants, if small changes in assumptions reverse every decision, or if the tape cannot reconcile to accounting, the market is not ready for autonomous capital allocation.

8 P34 and PARML as an Early Implementation

8.1 Profit-As-Regression Machine Learning

PARML is a model class trained directly against final economic outcomes rather than against a chain of intermediate forecasts. A classical enterprise stack might separately predict demand, price, conversion, return rate, default, shipping cost, and liquidation, then combine these estimates with handwritten rules. PARML can still use those signals, but its final learning target is the economic result of the action or portfolio.

For a menu M_t and candidate portfolio $p \subseteq M_t$, the model estimates

$$\widehat{Y}\left(p \mid I_{o,t}\right)$$

and selects a feasible portfolio according to the configured policy objective. The target can be portfolio profit, contribution margin, recovery, cash yield, liquidation value, or another final business measure. Learning the final outcome forces the system to internalize interactions among price, execution, timing, funding, risk, operating cost, and scale to the extent that those interactions are represented in the data.

Stable relationships discovered between telemetry and economic outcome are called **emergent business physics**. The term does not imply immutable natural law. It means empirical regularities that remain useful across perturbations, holdouts, forward periods, and live feedback. Such relationships should be probed, not anthropomorphized.

8.2 The core failure mode: optimistic models from biased history

P34 is designed around a common failure mode. Historical menus contain reliable profit mainly for actions that the prior policy selected. A powerful supervised regressor can therefore learn the economics of historically accepted deals while underestimating the loss rate among the wider unlabeled opportunity set. When market margins compress or the deployment policy expands beyond historical support, predicted positive actions can become a loss-making portfolio.

P34 treats **false-positive control** as a first-class problem. A false positive is not merely a row with sign error. It is an action that looks profitable under the learned historical pattern but carries hidden downside such as dead stock, failed execution, adverse selection, default, return, liquidation loss, or margin compression. In low-margin markets, avoiding false positives can matter more than finely ranking true positives.

8.3 Conceptual architecture

P34's current architecture can be understood as the following decision flow:

1. **Menu grounding.** Historical actions, sizes, labels, keys, dates, availability, and historical choices are validated and, where safe, expanded into grounded quantity alternatives.
2. **Regime scanning.** The system identifies changes in market difficulty, margin, liquidity, policy, demand quality, or signal reliability.
3. **Base nonparametric estimators.** Profit and false-positive telemetry are estimated using boosted and other flexible models over materialized representations.
4. **Universe construction.** Because unlabeled outcomes are not uniquely known, the system constructs multiple plausible interpretations of selection bias, unobserved opportunities, regime, and drift.
5. **Pareto filtering.** Universes dominated under calibration, downside, or policy objectives are removed.
6. **Controlled speculation.** Conservative and more opportunistic interpretations compete; speculative volume is accepted only when generalized telemetry supports it.
7. **Universe selection.** A meta-level selector chooses the interpretation most consistent with regime, portfolio behavior, calibration, false-positive telemetry, and the configured objective.
8. **Portfolio output.** The system returns executable key-quantity rows or no-trade, subject to constraints.

This structure is compatible with the MSD contract. The historical menu corresponds to market snapshots, historical choices correspond to `historically_taken`, and the sales tape provides a more explicit mechanism for reconstructing labels and open-position state than a single terminal-profit column.

8.4 Lifted labels and partial feedback

A practical system cannot rely on fully synthetic counterfactual labels. P34 uses **lifted labels**: holdouts, forward windows, out-of-fold predictions, portfolio calibration, false-positive telemetry, safe quantity grounding, and realized portfolio results. These signals supervise higher-level components even when the true outcome of every candidate row is unknown.

The universe selector is deliberately meta-level. It should not depend only on one market identity or one ticker. It uses generalized telemetry about how candidate portfolios behave. This is the bridge between a market-specific profit regressor and a cross-market computable-market model.

8.5 Engineering principles inherited from P34

The integrated architecture retains several design principles from the earlier paper:

- **Multi-lifted modularity.** Engineers can improve grounding, regimes, outcome models, universes, selectors, or optimization separately.
- **Boundary-aware sizing.** Zero, one unit, small quantities, and large discrete volumes are distinct regions; profit curves need not be monotonic.
- **Broad ingestion with auto-pruning.** New signals can enter quickly but must survive leakage, stability, and portfolio-value tests.
- **External signal reuse.** Forecasts and indicators from other markets can be added as menu features and judged by final economic value.
- **Human-understandable debugging.** Failures should be localized to menus, regimes, base labels, false-positive correction, universe choice, sizing, constraints, or external-signal drift.
- **No-trade as an action.** Waiting is a valid and often economically necessary output.

9 Synthetic Market Evidence

9.1 Benchmark purpose

The revised evidence uses the attached test_p34_09_slower_market_waves_demo notebook. It isolates selection bias under slow cyclical market dynamics, changing profitability, latent demand, downside regimes, operating costs, and a nonrandom historical acceptance policy with 1 percent exploration. Candidate deals are unrolled over quantity; the complete synthetic state is retained only for evaluation, while training views contain outcomes for business-selected or safely grounded actions.

The generator produced 3,931,200 candidate rows: 3,024,450 historical and 906,750 future-evaluation rows across 80 historical and 20 future dates. Each key had 39 quantity configurations. The historical set contained 77,550 keys; 13,367 were traded and had chosen-row outcomes, yielding 521,313 safely grounded quantity rows, or 17.24 percent of historical candidate rows. The P34 API payload used 3,024,450 historical menu rows, 22,443 sales rows, and a reserved week-80 task menu of 600 keys and 23,400 candidate rows with all task profits blank.

Observed and unobserved rows have different profit distributions, and the acceptance rate changes with market and policy state. The benchmark therefore tests whether an apparently strong model trained on the policy-filtered sample can safely expand to the full deployment menu. This is a synthetic diagnostic; it does not estimate a real market return distribution.

9.2 Baseline and API evaluation

The baseline is an XGBoost classifier trained on business-observed rows to predict profit above zero from lcogs, qty, sv, and isv. Chronological blocks were used for fit (6,202 rows), early stopping (2,660), calibration (2,660), and naive evaluation (4,505). The early-stop holdout achieved ROC-AUC 0.9266, PR-AUC 0.9467, and Brier 0.1003; the profit-calibrated cutoff was 0.819396.

The same fitted classifier and cutoff are evaluated twice: first on policy-observed weeks 64-79, then on all candidate rows in weeks 80-99 with one quantity retained per key. For P34, the same synthetic history is mapped to Menus/Sales with chosen-row historical profits, blank counterfactuals, and no outcomes on T=0. The default API model is scored on week 80 by joining returned key-quantity selections to hidden actuals.

The comparison is end-to-end, not feature-identical: XGBoost consumes four row features; P34 receives menus, a sales tape, unit economics, historical-choice indicators, a noisy feature, and market-type parameters. It should not be read as an isolated architecture ablation.

9.3 Results

Table 6. Conventional baseline: observed-data success and full-menu collapse.

Evaluation	Scored opportunity set	Model-selected actual profit	Take-All diagnostic	Take-None
Naive evaluation	Business-observed weeks 64-79	+$61,735.36	+$27,430.54	$0.00
Deploy-time simulation	Full future menu, weeks 80-99	-$417,370.54	-$521,713,014.88	$0.00

Table 7. Week-80 task-menu strategy comparison.

Strategy	Trades	Capital deployed	Predicted profit	Actual profit	Return on capital
P34 API portfolio	115	$4,145.72	+$2,404.07	+$2,249.52	54.26%
Calibrated XGB baseline	177	$18,820.53	n/a	-$8,788.91	-46.70%
Business policy (simulated)	180	$7,465.03	n/a	+$778.93	10.43%
Take-All diagnostic	23,400	$11,271,953.12	n/a	-$12,629,093.76	-112.04%
Take-None	0	$0.00	n/a	$0.00	n/a

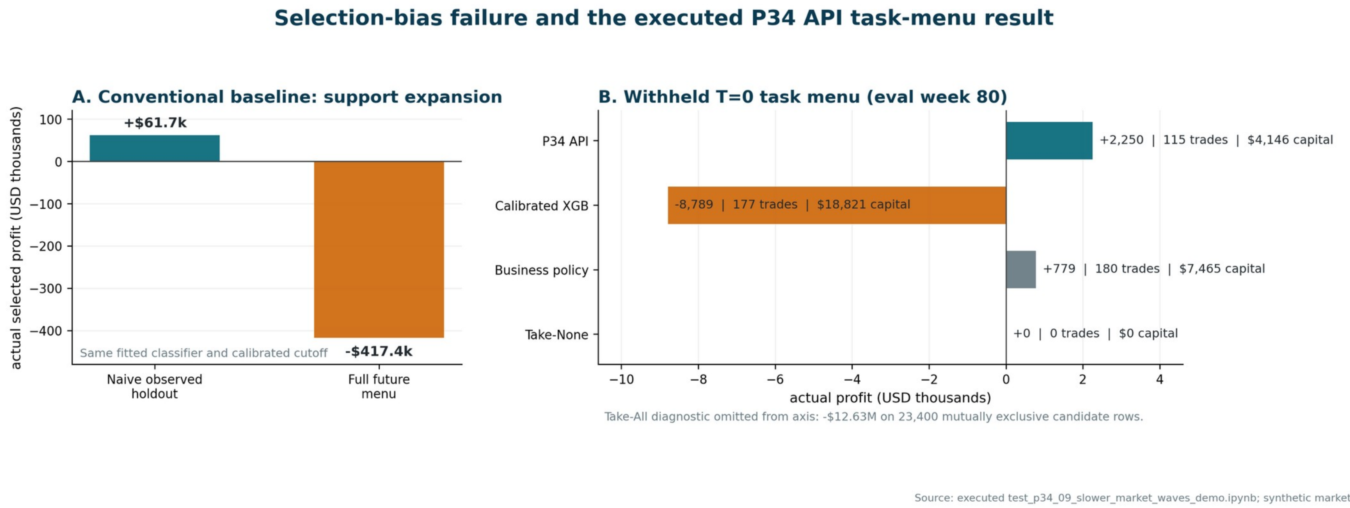

Figure 5. Selection-bias failure and the withheld task-menu result. Panel A shows the same calibrated XGBoost classifier earning +$61.7 thousand on the policy-observed holdout but losing $417.4 thousand on the full future menu. Panel B compares actual week-80 outcomes: P34 +$2,250, XGBoost -$8,789, historical business policy +$779, and no-trade $0; the Take-All diagnostic is omitted from the axis at -$12.63 million.

The baseline did not fail because its conventional holdout metrics were weak: ROC-AUC exceeded 0.92 and naive selected profit was positive. It failed when the evaluation support expanded from policy-filtered observations to the full opportunity menu, reversing the sign of selected profit. This is the specific false-positive collapse the synthetic experiment was constructed to expose.

On the withheld week-80 menu, P34 predicted +$2,404.07 and realized +$2,249.52, an aggregate actual-to-predicted ratio of 93.57 percent. Ninety-three of 115 selected trades were positive. P34 deployed $4,145.72, 22.03 percent of the XGBoost baseline's capital, and exceeded its realized profit by $11,038.43. These are one-menu synthetic results from an executed API call. They do not establish 20-week P34 performance, stationary-regime performance, causal identification, real-market profitability, or superiority under a feature-matched ablation.

10 Application Templates

The computable-market category is deliberately cross-industry. The following examples show how the same MSD roles map to different operating businesses. They are templates, not claims that every instance of each industry is immediately computable.

10.1 Amazon wholesale and retail inventory

A menu row can represent an ASIN, supplier, date, order quantity, acquisition cost, estimated selling price, platform fees, inventory state, sales velocity, competitor structure, lead time, and capital use. Quantity alternatives are separate actions because demand depth and liquidation risk are nonlinear.

The sales tape records purchase, inbound shipment, availability, unit sale, fee, return, reimbursement, removal, liquidation, and final closure events. Cash is extracted gradually. The business description defines FBA or fulfillment rules, shipping, advertising, return accounting, inventory aging, price policy, taxes, and shared overhead.

The main biases are historical deal selection, survivorship among suppliers and ASINs, and optimistic scaling from small quantities. The model's output is an order portfolio with quantities and a no-order option.

10.2 Microlending and credit-like operations

A menu row can represent an applicant, principal amount, term, price or APR, underwriting features, channel, acquisition cost, geography, collection capacity, and funding use. Different principal amounts are distinct actions.

The sales/event tape records disbursement, scheduled payment, actual payment, fee, delinquency, collection action, restructuring, default, recovery, write-off, and closure. The business description defines legal limits, affordability rules, servicing and collection processes, cost allocation, loss recognition, and risk appetite.

The historical policy signal is especially important because observed repayment outcomes come from approved applicants and offered amounts. Unobserved rejections are not random. Any model must also distinguish predictive performance from fair-lending and compliance obligations; governance can restrict features and decisions even where they appear profitable.

10.3 Customs timing and clearance insurance

A menu row can represent a shipment, route, commodity class, port, carrier, coverage amount, premium, deductible, expected clearance time, document quality, counterparty, and claim terms. The position is the contingent liability and service obligation created by underwriting or guaranteeing clearance timing.

The event tape records premium receipt, document events, inspection, delay, release, claim notice, payout, recovery, legal expense, and policy closure. The description defines the insured event, exclusions, claim adjudication, correlated exposure, legal jurisdiction, and operational interventions.

Computability depends heavily on attribution and correlated impact. A regulatory change or port disruption can affect many positions simultaneously, so portfolio dependence and scenario stress are essential.

10.4 Used-vehicle dealerships

A menu row can represent vehicle identity, auction, bid, transport, inspection, expected reconditioning, financing, listing channel, expected days-to-sale, lead environment, and size of any inventory package. The action includes the acquisition price and intended operational treatment.

The tape records purchase, transport, repair, financing cost, listing, lead, test drive, price change, trade-in, sale, warranty expense, return, and settlement. The business description defines reconditioning standards, sales commissions, floor-plan financing, title and regulatory processes, aging policy, and liquidation channels.

Operator-dependent value is obvious: two dealerships can have different repair costs, local demand, financing, and conversion. A market-average valuation is not the same as operator-relative profit.

10.5 Freight and load boards

A menu row can represent lane, date, equipment, bid, expected fuel and labor, deadhead, counterparty, service window, and capacity. The tape records acceptance, pickup, delay, accessorial charge, delivery, claim, payment, and closure. Menus may be virtual rolling windows because offers arrive continuously rather than in synchronized batches.

10.6 GPU and compute-capacity allocation

A menu row can represent workload, customer, hardware type, duration, price, energy, preemption, service-level commitment, data locality, and opportunity cost. The tape records reservation, start, utilization, failure, migration, completion, credit, energy cost, and payment. The model selects workloads and capacity blocks under thermal, energy, hardware, and contractual constraints.

10.7 Thin digital assets and alternative trading venues

A menu row must include asset, size, venue, order type, timing, available depth, volatility, funding, and exit policy. The tape records fills, fees, transfers, partial liquidations, slippage, and closure. Displayed prices are insufficient when the intended size exceeds visible depth. Market-impact and strategic-adversary modeling may dominate all other components.

Table 8. Cross-market mapping of the MSD contract.

Market	Menu snapshot	Sales / event tape	Description emphasis	Primary computability risk
Wholesale inventory	SKU-supplier-size-price alternatives	purchases, unit sales, returns, fees, liquidation	fulfillment, pricing, aging, shared costs	demand depth and optimistic scaling
Microlending	applicant-amount-term-price alternatives	disbursement, payments, delinquency, recovery	underwriting, collections, legal limits	selection bias and delayed default
Clearance insurance	shipment-coverage-premium choices	premium, delay, claim, payout, release	policy terms, correlated events	systemic shocks and claims attribution
Used vehicles	vehicle-bid-reconditioning-channel choices	purchase, repair, leads, sale, settlement	financing, operations, title, aging	heterogeneous condition and local demand
Freight	lane-equipment-bid-capacity choices	accept, pickup, delivery, claims, payment	service rules and capacity	rapidly expiring menus and interference
GPU capacity	workload-hardware-duration-price choices	reserve, run, fail, complete, bill	SLA, energy, preemption	nonstationary utilization and opportunity cost
Thin assets	venue-size-order-exit choices	fills, fees, transfers, liquidation	execution and custody	support, impact, manipulation, latency

11 Evaluation and Falsification Protocol

11.1 Evaluation must follow decision time

Random row splits are generally inappropriate because they mix regimes, leak near-duplicate actions, and preserve the historical policy distribution. A computable-market evaluation should reconstruct what was known at each decision time and evaluate complete menus in forward order.

At minimum, the protocol should include:

1. **date-aware historical splits** with no post-decision leakage;
2. **menu-level evaluation**, not independent-row metrics only;
3. **portfolio reconstruction** using the exact constraints and mutual-exclusion rules in force;
4. **outcome maturity rules** for open and delayed positions;
5. **baseline comparison** to the current business policy and strong market-specific models;
6. **support reporting** for every recommendation;
7. **stress tests** for margin compression, liquidity shock, missing labels, correlated downside, and policy change;
8. **shadow deployment** before live capital; and
9. **independent reconciliation** of selected actions to realized event tape and accounting.

11.2 Metrics

Prediction metrics remain useful, but the primary metrics should reflect decisions.

Table 9. Recommended computable-market evaluation metrics.

Metric	Purpose
Realized portfolio profit or utility	Measures the economic result of executed or reconstructed selections.
Predicted-to-realized calibration	Detects systematic optimism or conservatism.
Positive-portfolio rate	Measures how often complete decision periods finish positive.
False-positive exposure	Capital or quantity allocated to actions predicted positive but realized negative.
Downside quantiles and expected shortfall	Evaluates tail protection and risk policy.
Regret versus baseline policy	Measures improvement over current business behavior under comparable constraints.

Metric	Purpose
Oracle efficiency in synthetic or randomized settings	Measures captured value when counterfactual outcomes are available.
Capital turnover and days-to-cash	Captures speed of base-asset recovery.
Support-weighted performance	Separates grounded decisions from extrapolative ones.
Constraint and compliance violations	Measures executability and governance failure.
Decision latency and compute cost	Tests whether the model can operate before menus expire.
Cross-market transfer gain	Compares pretrained adaptation to training from scratch.

11.3 Off-policy evaluation and exploration

Historical evaluation is difficult because outcomes are observed mainly for actions selected by the logging policy. Propensity methods, doubly robust estimators, conservative offline learning, and partial-identification bounds can help, but they depend on support and modeling assumptions. When permissible, small randomized or structured exploration is valuable because it turns otherwise unidentifiable regions into evidence.

Exploration must be budgeted as a risk-bearing investment. The system should record exploration probability, reason, exposure, and outcome. Randomization is not appropriate where legal, ethical, or catastrophic downside prohibits it. In those markets, safe simulators, natural experiments, domain constraints, and conservative support boundaries become more important.

11.4 Cross-market pretraining tests

The central cross-market hypothesis should be evaluated through held-out-market designs:

- pretrain on a set of markets and withhold an entire industry or operator;
- compare zero-shot, few-shot, and fully adapted performance;
- permute column names to test semantic dependence;
- remove the business description to measure its marginal value;
- remove the sales tape or aggregate it to test path dependence;
- remove `historically_taken` to measure policy-bias correction;
- introduce unseen cost rules, size curves, and delays;
- test whether transfer helps in small data but yields to local evidence as data grows.

A publishable claim of a universal market model requires these harder tests, not only within-market holdouts.

11.5 Falsification criteria

The computable-market thesis is weakened for a proposed market when repeated controlled studies show any of the following:

- no model consistently improves on simple baselines after accounting and constraints;
- simulated universes cannot reproduce observed event-tape invariants;
- performance disappears when post-decision leakage is removed;
- recommended actions depend primarily on unsupported regions;
- small changes in the business description reverse decisions without empirical resolution;
- market impact created by the policy invalidates historical relationships faster than retraining can respond;
- capital-adjusted gains are smaller than data, compute, integration, and governance cost; or
- live shadow results remain uncalibrated despite sufficient outcome maturity.

The concept is scientific only if individual markets can fail the computability test.

12 Safety, Governance, and Deployment

12.1 Profit-directed automation is a high-consequence system

A model that directly allocates capital, inventory, credit, insurance exposure, or operational capacity can create material financial and legal consequences. A computable-market system should be governed as a decision system, not deployed as an ordinary dashboard. The fact that the objective is profit does not make every profitable-looking action permissible, and the fact that a policy is statistically calibrated does not make it operationally executable.

The recommended deployment path is staged:

1. data and accounting reconciliation;
2. retrospective replay with date-aware controls;
3. synthetic stress testing;
4. independent methodology review;
5. live shadow recommendations with no automatic execution;
6. limited capital or capacity exposure;
7. monitored expansion by market, action type, and support level; and
8. continuous rollback and kill-switch readiness.

12.2 Primary risks and controls

Table 10. Primary deployment risks and mitigations.

Risk	Failure mode	Required controls
False-positive over-entry	The system selects many actions that appear profitable but hide downside.	downside telemetry, conservative universes, exposure caps, no-trade defaults, shadow calibration
Distribution shift	Margin, demand, policy, or cost structure changes after training.	regime detection, time-weighting, drift alarms, forward calibration, rollback rules
Unsupported transfer	Cross-market priors dominate weak local evidence.	support scores, local-vs-prior attribution, abstention, staged adaptation
Market impact and interference	Executing one action changes the value or availability of others.	impact models, concentration limits, sequential re-pricing, market-specific guardrails
Outcome misattribution	Cash flows and costs are linked to the wrong position or allocated arbitrarily.	identity lineage, accounting reconciliation, closure tests, audit sampling
Data leakage	Future or post-decision information enters features.	feature provenance, availability timestamps, replayable pipelines, date-aware splits
LLM simulation error	Plausible text produces false rules, arithmetic, or causal claims.	tool execution, deterministic code, tests, provenance, universe weighting, human review
Automation failure	APIs, inventory, payment, servicing, or execution systems fail.	throttles, idempotency, reconciliation, manual override, retry and cancellation policy
Regulatory or contractual breach	Automated decisions violate law, platform terms, or counterparty restrictions.	constraint compilation, allowed-action lists, compliance sign-off, immutable audit logs
Adversarial response	Counterparties or competitors react to the model or manipulate telemetry.	anomaly detection, exposure limits, adversary testing, execution diversity
Objective misspecification	The configured profit target omits risk, capital, customer, or long-run effects.	objective review, multi-metric dashboard, board-level risk appetite, periodic red-team audit

Risk	Failure mode	Required controls
Policy inheritance	historically_taken causes the system to reproduce obsolete or harmful behavior.	treat behavior as prior, not truth; fairness and compliance tests; explicit policy overrides

12.3 Governance artifacts

Each live decision should be replayable from a versioned bundle containing the menu, event-tape cutoff, business description, model version, constraints, simulation universes, selected policy, support diagnostics, and output. The system should expose at least:

- predicted and realized portfolio economics;
- uncertainty and false-positive warnings;
- current regime and drift indicators;
- concentration and capital use;
- support and extrapolation level;
- universe or simulator influence;
- constraint binding and reason codes;
- execution reconciliation; and
- kill-switch status.

Independent auditors should be able to reproduce labels, splits, portfolio construction, and reported benchmark values without access to production secrets that are irrelevant to validation.

13 Positioning Relative to Existing Systems

Computable-market decisioning does not map cleanly to one current software category. Its closest substitutes are assembled stacks of portfolio analytics, AutoML, forecasting, optimization, offline policy learning, decision orchestration, and custom integration.

Table 11. Adjacent categories and the computable-market distinction.

Category	Strength	Gap relative to the proposed category
Portfolio and risk platforms	Whole-portfolio accounting, risk, compliance, and workflows	Usually assume defined instruments and models rather than learning operator-specific profit from biased business menus.
Quant research and execution infrastructure	Backtesting, strategy coding, broker connectivity, optimization	Generally expects the user to specify the strategy and works best with standardized market data.
AutoML and tabular ML	High-quality supervised prediction from a defined table	Does not by itself define menus, reconstruct event outcomes, correct missing counterfactuals, or output feasible portfolios.
Decision intelligence and rules	Operational orchestration, approvals, and policy management	Usually combines rules and narrow models rather than learning a general menu-to-portfolio profit policy.
Offline RL and contextual bandits	Policy learning from logged partial feedback	Requires careful support, propensity, dynamics, and safety assumptions; business interface and accounting remain unsolved.
Table foundation models	Reusable tabular priors and semantic transfer	Primarily predict labels; do not natively model event tapes, portfolio constraints, or economic execution.

Category	Strength	Gap relative to the proposed category
Financial language/time-series models	Language understanding, market research, or sequence forecasting	Can supply features but do not define operator-relative menu-to-cash decisioning.
Optimization solvers	Exact or approximate solution of constrained allocation problems	Require objective coefficients and dynamics to be supplied; they do not learn them from biased telemetry.

P34 is best understood as an early **decision layer between enterprise telemetry and executable economic action**. AutoML, offline RL, LLMs, simulators, and solvers can all be components inside this layer.

14 Market Implications: From Hidden Telemetry to Asset-Like Value

14.1 Latent-space speculation

Many operational markets contain economic state that is real but not published as a standardized market feed. Inventory is in transit or processing; loan quality is dispersed across servicing events; vehicle condition is partly tacit; freight capacity changes minute by minute; clearance risk is embedded in documents and workflows. The market signal exists in latent form across operational telemetry.

Latent-space speculation is the estimation of economic value in markets where the relevant state is not explicitly materialized as a clean order book or standardized data product. The term does not imply ungrounded gambling. It describes inference over hidden but partially measurable operational state.

14.2 Latent-space recovery

When menus, event tapes, and descriptions are reconciled, a business can convert hidden operations into structured expected-value estimates, risk measures, and auditable performance. This process is **latent-space recovery**. It can improve internal decisions, but it may also change the financial character of operational assets.

A purchase order, manufacturing batch, lead stream, capacity block, receivable, or shipment guarantee is not necessarily a security. Nevertheless, it can have measurable expected cash flow, duration, downside, and liquidation value. If predictions become sufficiently calibrated and independently verified, they may support financing, risk transfer, secondary sale, or benchmark valuation. The analogy to standardized option pricing is organizational rather than mechanistic: shared valuation methods can make previously opaque claims more comparable and liquid.

Any move from internal decisioning to financing or resale raises separate securities, lending, insurance, accounting, and disclosure questions. A model's output does not determine legal classification. Market creation must proceed with jurisdiction-specific counsel and independent risk governance.

14.3 Information-processing scale as competitive advantage

The deepest economic advantage in a computable market may be **computational comparative advantage**: the ability to evaluate a larger and more heterogeneous opportunity set at transaction-level resolution without accepting lower-quality marginal deals. A business with the same suppliers, customers, and capital as its competitors can earn different returns if it has better telemetry, lower decision latency, more accurate sizing, and stronger false-positive control.

This advantage can be self-reinforcing. Better selection creates cleaner event histories; better histories improve models; better models support larger volume and more structured exploration; and larger but controlled volume produces more evidence. The loop can also fail self-reinforcingly if early models over-enter, corrupt labels through stressed operations, and learn from their own degraded execution. Governance and capacity modeling determine which path dominates.

15 Limitations and Research Agenda

15.1 Current limitations

The concept and P34 evidence are early. First-order limitations include:

- **Synthetic benchmark evidence.** The principal reported results use synthetic, real-world-looking markets. They isolate known failure modes but do not prove live performance.
- **Incomplete cross-market validation.** A fully pre-trained and pre-aligned universal market decision model has not yet been demonstrated across the breadth claimed by the thesis.
- **Limited regime coverage.** The P34 selector has been evaluated on a restricted library of regimes compared with real operational diversity.
- **Not fully differentiable.** Current P34 components use boosted models and materialized matrices in several blocks. This aids debugging but does not realize a fully end-to-end architecture.
- **Portfolio unrolling cost.** Large menus and many size/timing configurations can create severe computational growth.
- **Menu quality dependence.** A model cannot reliably optimize actions that are misrepresented, unavailable, or missing critical constraints.
- **Low-liquidity and interference difficulty.** If one action changes many others in an unobserved way, additional impact and sequential-decision modeling is required.
- **Distant-menu detection.** Automatic abstention when a current menu is far from training context remains an essential development item.
- **Feedback delay.** Long-maturity markets can change before enough outcomes arrive for recalibration.
- **Description ambiguity.** Natural-language rules can conflict, omit exceptions, or differ from actual operations.
- **Identity and cost allocation.** Shared expenses, partial positions, returns, refinancings, and merged workflows can make profit attribution contestable.
- **Latency.** Depending on menu size, universe budget, and simulation depth, decisions may take seconds, minutes, or hours; some high-frequency markets are outside scope.
- **Manipulation and performativity.** Competitors, counterparties, or internal teams may react to the policy, invalidating passive historical assumptions.
- **Objective risk.** Direct profit optimization can externalize customer, employee, regulatory, reputational, or systemic harms unless these enter constraints and utility.
- **No profit guarantee.** Adversarial shocks, discontinuities, execution failures, and invalid data can defeat any learned policy.

15.2 Research priorities

A credible research program should focus on the following milestones.

Table 12. Priority research and engineering milestones.

Milestone	Research question or deliverable
Public benchmark suite	Can different methods handle selection bias, delayed feedback, liquidity shocks, drift, sparse labels, and correlated downside on common menu-tape tasks?
Held-out-market transfer	Does cross-market pretraining improve few-shot decisions on entirely unseen industries and operators?
Computability score	Can instrumentation quality, support, feedback delay, impact, and signal stability predict whether a market is worth modeling?
Differentiable multi-lifted architecture	Which stages benefit from end-to-end gradients, and which should remain explicit, modular, or solver-based?
Simulation governance	How should LLM-generated rules and pseudo-labels be validated, weighted, versioned, and retired?
Support-aware no-trade	Can the model reliably abstain when current menus are distant from local and pretrained evidence?

Milestone	Research question or deliverable
Policy-alignment theory	How should historically_taken encode risk appetite without reproducing obsolete or unlawful behavior?
Sequential market impact	How should the system reprice remaining menu options after its own actions change state and competition?
Auditable emergent physics	Can stable learned relationships be expressed as testable economic hypotheses rather than opaque feature importance?
External signal marketplace	Can third-party forecasts and models be added under a standard provenance, evaluation, and compensation protocol?
Independent live trials	Do synthetic robustness gains survive shadow and staged live deployment with full accounting?

15.3 Strong claims that remain unproven

Three claims should be treated as hypotheses until broad evidence exists.

First, **two tables plus a business description are universally sufficient in practice**. The paper defends them as a universal interface, but some markets may require additional modalities such as images, documents, geospatial streams, or order-book sequences. These can be referenced by menu or event rows, but the engineering boundary may need extension.

Second, **one pre-trained market model can transfer across radically different industries**. Table-semantic and prior-data-fitted models support plausibility, not proof. Economic transfer may depend more on shared causal and accounting structure than on superficial table similarity.

Third, **more engineering effort and compute monotonically improve decisions**. They improve the frontier only when used to reduce a real error term and paired with valid information. More compute can also search more aggressively into model error, amplify leakage, and create more confident false positives.

16 Conclusion

A computable market is an economic environment in which a particular operator can represent feasible actions, reconstruct realized outcomes, express operating constraints, and improve a policy through repeated feedback. The category is most relevant where opportunity sets are large, positive expected value is sparse, returns are operator-dependent, size and execution are nonlinear, labels are selected by historical policy, and decisions must be made at a frequency or precision beyond manual analysis.

The proposed Menu-Sales-Description contract gives this category a portable systems boundary. The menu is the market snapshot. The sales table is the event tape through which positions return to the base asset. The business description is the operational constitution that defines accounting, workflow, costs, constraints, and market laws. A current menu configures the next decision. historically_taken exposes the prior policy and its risk appetite without pretending that the old policy is ground truth.

The Computable-Market Thesis is an information-relative claim. Under adequate support, attribution, feedback, stability, and impact modeling, a sufficiently expressive, pre-trained, and locally aligned nonparametric decision system should be able to reduce policy regret toward the best use of the information encoded in the contract. The thesis does not imply perfect prediction or guaranteed profit. It states that a large class of operational capital-allocation problems can be transformed from tacit judgment and brittle rules into measurable, testable, and progressively improvable decision systems.

P34/PARML is an early implementation. The revised notebook shows a strong conventional classifier succeeding on policy-filtered history and collapsing on the full synthetic menu. On the withheld week-80 API test, P34 produced a positive, closely calibrated portfolio while the baseline lost money. Because the market is synthetic, only one P34 menu is tested, and the interfaces are not feature-identical, the result is diagnostic rather than proof of broad or live performance. The larger research agenda is to identify computable markets, test cross-market transfer, ground simulation safely, and audit autonomous decisions before scale.

Appendix A. Formal Definitions and Results

A.1 Definition A1: Business menu

A business menu M_t is a finite or countable set of candidate state-action configurations available to operator o at decision time t, together with decision-time features, identities, quantities, constraints, outcome observability, and historical-policy indicators where applicable.

A.2 Definition A2: Sales/event tape

A sales tape $S_{\le t}$ is a temporally ordered, replayable collection of events keyed to positions or actions, sufficient under the business accounting rules to reconstruct changes in exposure, cash, costs, status, and closure.

A.3 Definition A3: Business description

A business description B_o is a versioned operational constitution specifying the economic objective, asset ontology, workflow, costs, constraints, time conventions, market-impact assumptions, data semantics, prior policy, and governance rules for operator o.

A.4 Definition A4: Computable market

A market M is computable for operator o to tolerance $\left(\epsilon, \delta, T\right)$ under contract I if a feasible policy can be learned and evaluated such that information-relative regret over horizon T is at most ϵ with probability at least $1-\delta$, under the stated distribution, controls, and outcome-maturity assumptions.

A.5 Definition A5: PARML

Profit-As-Regression Machine Learning is a model class that estimates and optimizes final economic outcomes of actions or portfolios directly, while allowing intermediate forecasts, simulations, and rules to enter as features or structural components.

A.6 Result A1: Information monotonicity

Let information set I_1 be measurable with respect to richer information set I_2. If the feasible policy class under I_2 includes policies that ignore the additional information, then the optimal expected utility under I_2 is weakly greater than under I_1. This is a decision-theoretic property of the optimum, not a guarantee that a finite-sample learning algorithm improves when given noisy features.

A.7 Result A2: Static plug-in policy consistency

Under bounded utility, adequate support, consistent outcome estimation, and a finite feasible action set, uniform convergence of estimated conditional utility implies convergence of the value of the plug-in argmax policy to the information-optimal value. Portfolio extensions require uniform control over the feasible portfolio class and continuity or appropriate combinatorial generalization bounds.

A.8 Dynamic caveat

When actions affect future states, the event tape and menu sequence define a partially observed controlled process. The static result does not identify dynamic counterfactuals without additional assumptions. A practical architecture therefore combines conservative support, explicit simulators, regime models, sequential re-evaluation, and staged live feedback.

Appendix B. Canonical Interface Examples

B.1 Example B1: Historical menu rows

Table B1. Simplified historical menu.

menu_id	decision_time	asset_id	size	acquisition_cost	expected_price	feature_1	historically_taken	profit	label_provenance
2026-05-01-A	2026-05-01	ASSET-101	10	420	52	0.81	1	63.40	observed
2026-05-01-A	2026-05-01	ASSET-101	25	1,050	52	0.81	0	NaN	unobserved
2026-05-01-A	2026-05-01	ASSET-204	5	310	79	0.44	0	NaN	unobserved
2026-05-08-B	2026-05-08	ASSET-305	12	960	91	0.67	1	-84.10	observed

The rows for ASSET-101 are mutually exclusive size configurations. The missing profit in the unchosen 25-unit row is not zero. The 10-unit row can potentially ground smaller quantities if the business rules and event order support that inference.

B.2 Example B2: Sales/event tape

Table B2. Simplified event tape.

event_id	position_id	origin_menu_id	event_time	event_type	quantity_delta	cash_flow_base	status_after
E-001	P-101	2026-05-01-A	2026-05-01	open/purchase	+10	-420.00	open
E-002	P-101	2026-05-01-A	2026-05-06	sale	-4	+208.00	partial
E-003	P-101	2026-05-01-A	2026-05-12	sale	-6	+312.00	closed
E-004	P-101	2026-05-01-A	2026-05-12	fee	0	-24.60	closed
E-005	P-101	2026-05-01-A	2026-05-12	allocated_operation_cost	0	-12.00	closed

The reconstructed profit is $520.00 - $420.00 - $24.60 - $12.00 = $63.40.

B.3 Example B3: Current menu and output

Table B3. Current menu input.

menu_id	decision_time	asset_id	size	acquisition_cost	expected_price	feature_1	historically_taken	profit
2026-08-08-C	2026-08-08	ASSET-101	8	344	54	0.76	NaN	NaN
2026-08-08-C	2026-08-08	ASSET-101	20	860	54	0.76	NaN	NaN
2026-08-08-C	2026-08-08	ASSET-410	6	570	109	0.59	NaN	NaN

Table B4. Decision output.

asset_id	candidate_size	selected	selected_size	predicted_profit	10th-percentile profit	support_score	reason_code
ASSET-101	8	1	8	41.20	7.10	0.88	selected_supported
ASSET-101	20	0	0	26.00	-74.00	0.42	capacity_false_positive
ASSET-410	6	0	0	-9.50	-51.00	0.67	no_trade_negative_ev

Appendix C. P34 Benchmark Methodology

The revised notebook benchmark follows these steps:

1. Generate a slower-wave synthetic inventory market with 80 historical dates, 20 future dates, and 39 quantity rows per key.
2. Apply a biased historical acceptance policy with 1 percent exploration; retain the complete synthetic state only for evaluation.
3. Convert the market to business-observed history and train an XGBoost classifier on the positive-profit label using chronological fit, early-stop, calibration, and naive-evaluation blocks.
4. Calibrate the classifier threshold against realized profit, then evaluate it on the final policy-observed historical weeks.
5. Apply the same fitted classifier and threshold to the full future menus for weeks 80-99.
6. Build the P34 Menus and Sales sheets from the full historical menu, chosen-row profits, blank counterfactual outcomes, replayed sales, and synthetic-inventory market parameters.
7. Reserve week 80 as the T=0 task menu: 600 keys, 23,400 quantity rows, and no task outcomes in the request.
8. Submit or re-attach to the P34 REST fit session, poll the result, and treat absent keys or qty = 0 as no-trade decisions.
9. Join P34 selections to hidden week-80 actuals and compare them with the calibrated XGBoost baseline, simulated business policy, Take-All diagnostic, and Take-None.
10. Report actual profit, predicted P34 profit, deployed capital, return on capital, positive selected trades, and aggregate calibration.

The Take-All row is a non-deployable diagnostic that sums every candidate quantity row, including mutually exclusive size alternatives. The notebook evaluates P34 on one task menu; it does not provide the prior paper's two-scenario aggregate P34 benchmark or an oracle-efficiency estimate. Accordingly, those earlier headline figures have been removed from the revised results section.

Appendix D. Glossary

Table D1. Core terms.

Term	Definition
Computable market	An operator-relative market in which actions, outcomes, constraints, and repeated feedback can support measurable policy improvement.
MSD contract	The Menu-Sales-Description interface: opportunity snapshots, realization event tape, and operational constitution.
Business menu	Structured set of candidate actions available at a decision time.
Sales/event tape	Keyed history of position realization, cash extraction, costs, status, and closure.
historically_taken	Behavior-policy indicator showing what the prior business selected.
PARML	Profit-As-Regression Machine Learning, trained toward final economic outcomes.
P34	HyperC’s current PARML decision architecture.
Safe grounding	Label expansion justified by nested quantities, accounting identities, or verified operational relations.
False positive	An action predicted profitable that realizes negative economics or unacceptable downside.
Universe	A plausible interpretation of unobserved outcomes, selection bias, regime, drift, and impact.
Universe selector	Meta-level component choosing among plausible universes under policy objectives.
Latent-space speculation	Estimating value where relevant market state is real but not explicitly materialized as clean market data.
Latent-space recovery	Converting hidden operational telemetry into structured economic signal, auditability, and decision infrastructure.

Term	Definition
Emergent business physics	Stable empirical relationships between telemetry and economic outcomes that survive validation and perturbation.
Information-relative optimum	Best feasible policy measurable from the supplied information, not an omniscient market optimum.
Support	Evidence that a contemplated action or state is represented by historical, grounded, or defensible simulated experience.

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